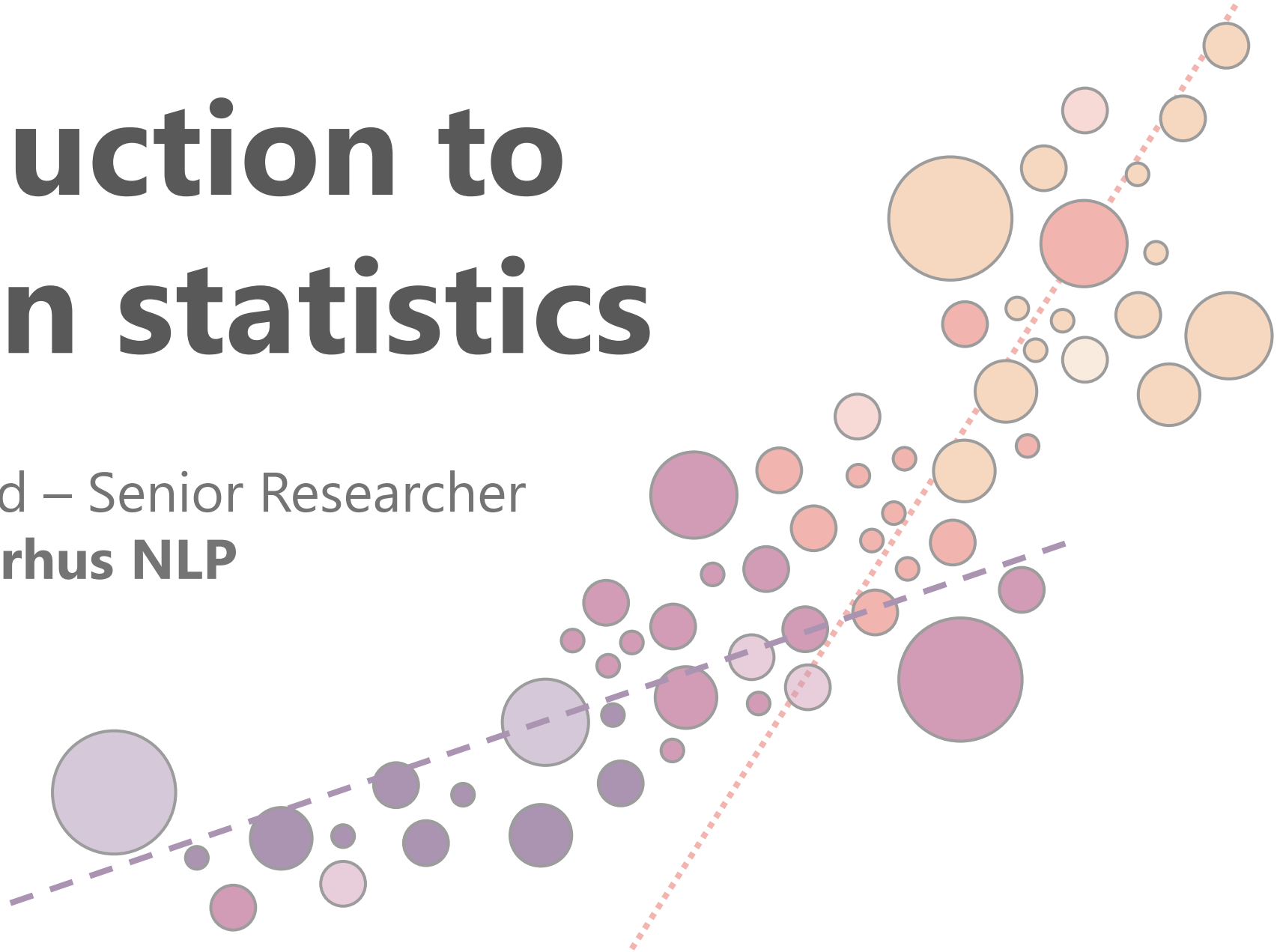
Introduction to Bayesian statistics

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Two tales on probability

But wait… what is a probability?

What is the probability that a coin flip comes up heads?

What is the probability that it rained in London on January 1st, 1750?

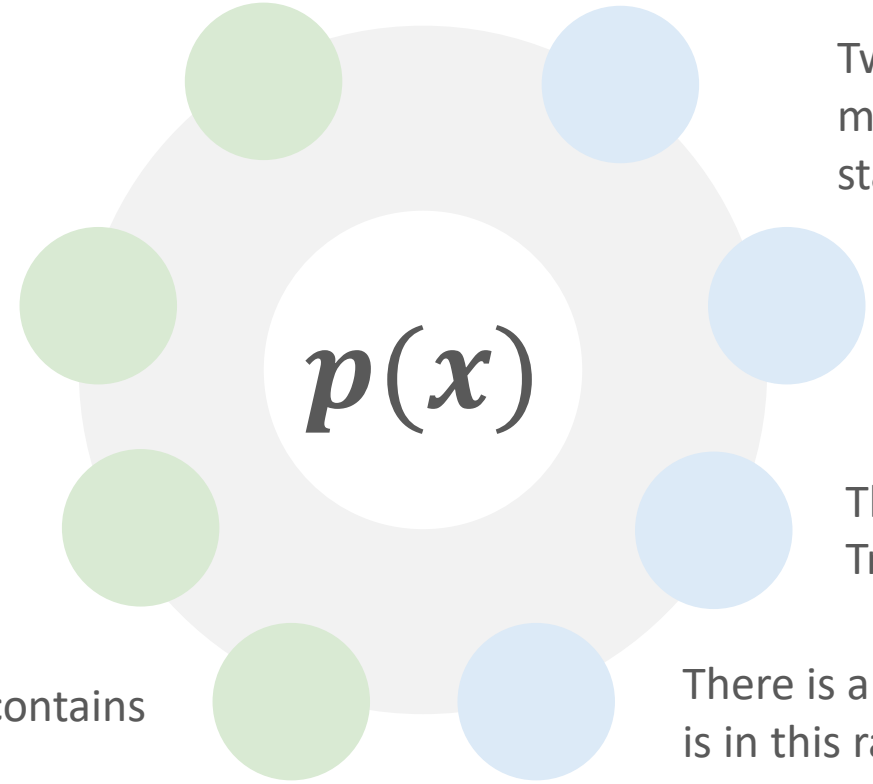
Frequentism

Two researchers looking at the same data should ideally arrive at the **same p-value**

Probability is the limit of a **long-run frequency**

The data is a **random** sample of a fixed truth

I am 95% confident the range contains the truth (**Confidence interval**)



Bayesianism

Two researchers looking at the same data might reach **different conclusions** if they started with different priors

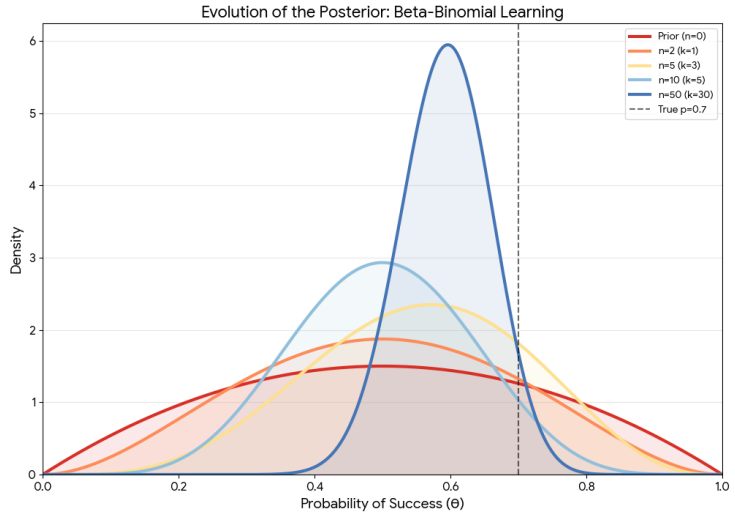
Probability is a **degree of belief** or certainty.

The data is **fixed**, and the Truth is what is random.

There is a 95% probability that the truth is in this range (**Credible interval**).

The Iterative Nature of Learning

Today's Posterior is tomorrow's Prior

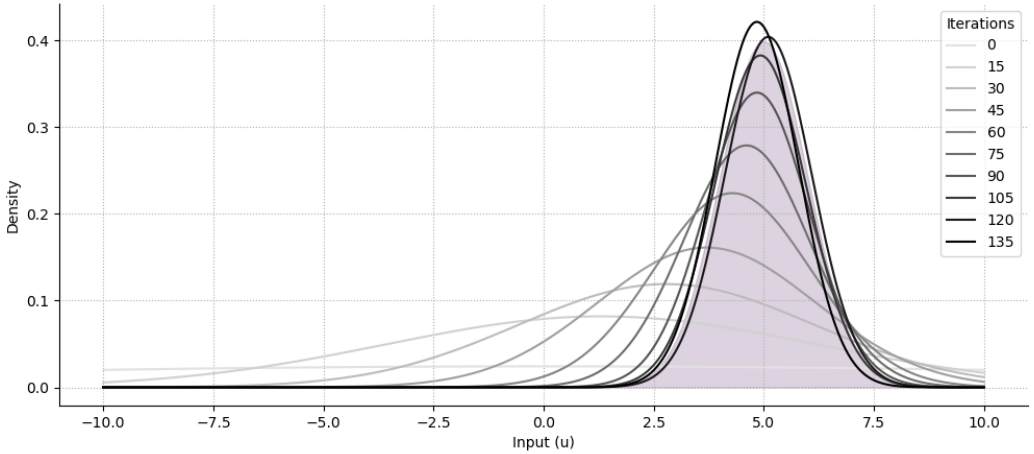


Forward Probability (Frequentist)

You have a jar with 70 red marbles and 30 blue marbles. If you reach in, what is the chance you pull a red one?

Inverse Probability (Bayesian)

You pull out 3 red marbles. Now, what is the probability that the jar *originally* had 70% red marbles?



The Anatomy of Bayes' Theorem

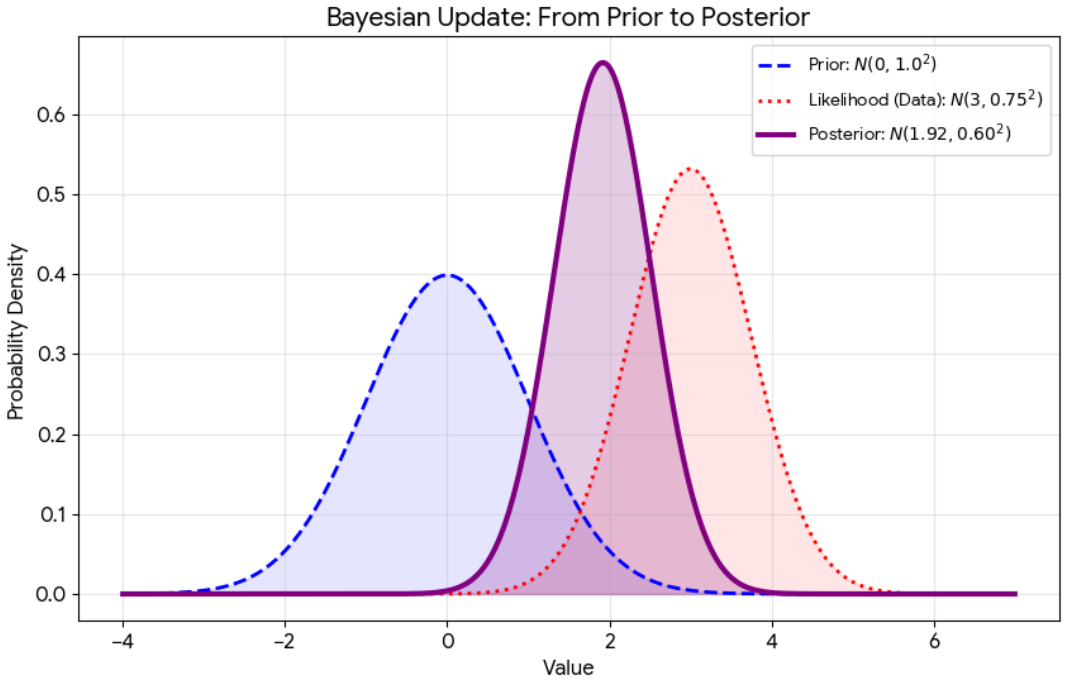
Likelihood Prior

$$P(\theta|y) = \frac{P(y|\theta)P(\theta)}{P(y)}$$

Marginal probability

$$\int P(y|\theta)P(\theta)d\theta$$





How do we get there?

Until the mid 1990s, that was impossible.



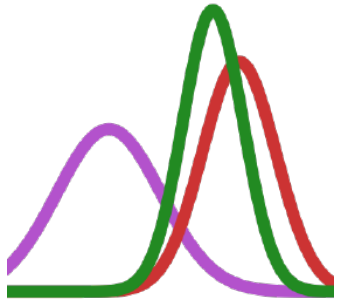


Frequentist methods (like Maximum Likelihood Estimation) are often **optimization problems** (finding the peak of a curve). This is usually fast.





Bayesian methods are often **integration problems** (finding the area under a curve). This is computationally expensive and usually requires simulation techniques like **MCMC (Markov Chain Monte Carlo)**.





Markov Chain Mont Carlo

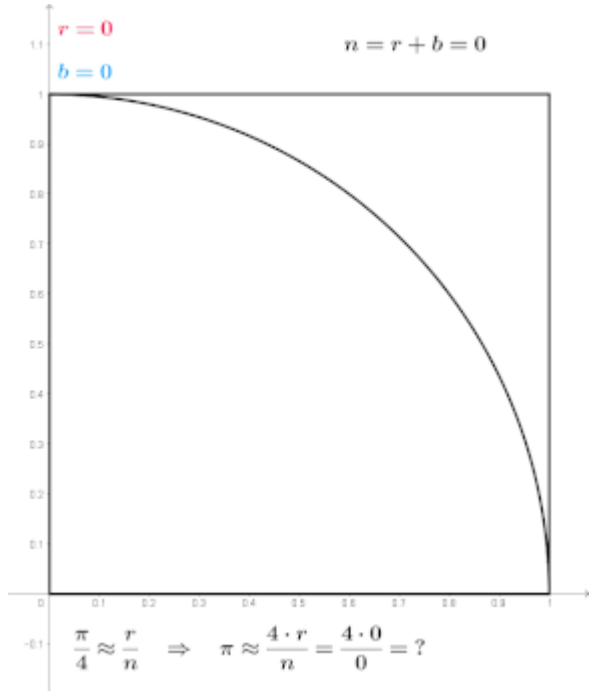
$$P(X_{n+1} = x | X_1 = x_1, X_2 = x_2, \dots, X_n = x_n) = P(X_{n+1} = x | X_n = x_n)$$

See how it works here.



Monte Carlo method

Use randomness to solve problems that might be deterministic in principle.

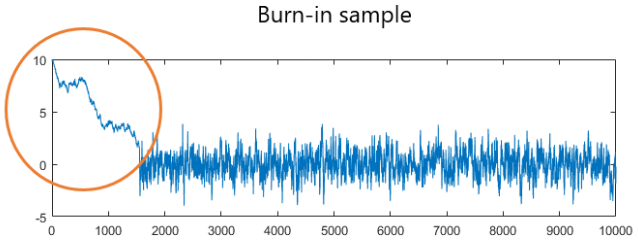


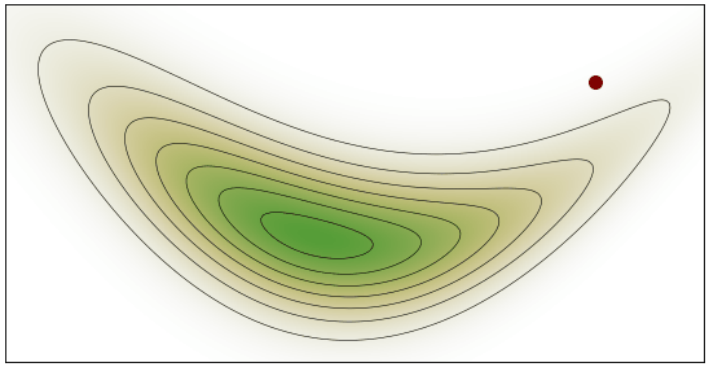
Gibbs sampling

Metropolis-Hasting algorithm

Hamiltonian Monte Carlo

The No U-Turn Sampler (NUTS)





See how it works here.